Does Poverty Affect Life Expectancy? A Study on US County Poverty and Life Expectancy while Accounting for Education and Urbaneness (2015)

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Question	For 2015 US counties, does a higher poverty rate independently predict lower life expectancy at birth (years) after controlling for adult education and metro status?
Hypothesis	Higher poverty should shorten life through less access to care, poorer nutrition and housing, and chronic stress.
Headline finding	After controlling for adult education and metro status, a county with a one-percentage-point higher poverty rate has about 0.17 fewer years of life expectancy (95% CI: −0.189 to −0.150; N = 3,108 counties). This is a conditional association, not a causal effect: unmeasured health, demographic and regional factors plausibly move poverty and longevity together, and causation likely runs in both directions.
Data	CDC NCHS USALEEP tract life expectancy, aggregated to counties with 2010 Census tract populations; Census SAIPE 2015 poverty rates; USDA ERS adult education (2008–12) and urban–rural codes. One row is one US county, final N = 3,108.
Method	OLS of county life expectancy on the poverty rate, the share of adults without a high school diploma and a metro indicator, with HC3 heteroskedasticity-robust standard errors.

1. The Question and the Data

Life expectancy at birth differs by more than twenty years across US counties, from 68.9 to 89.5 years in the sample analyzed here. A central question for public health research, and for the design of anti-poverty and health policy, is whether that variation reflects material deprivation itself or the characteristics of the places and people that deprivation tends to accompany. Poverty is correlated with low educational attainment and with rural residence, and both are independently associated with health. If poverty remains associated with shorter lives among counties that share the same educational profile and the same degree of urbanization, then deprivation is a health problem in its own right and anti-poverty policy is plausibly health policy. If the association disappears once those characteristics are held fixed, the poverty–health gradient is better understood as a story about who lives where and who is educated. The question is therefore relevant to public health researchers, to the agencies that allocate federal funds based on poverty estimates, and to local policymakers deciding where health investments will matter most.

Three datasets were used. County life expectancy is built from the U.S. Small-area Life Expectancy Estimates Project (USALEEP), produced by the CDC's National Center for Health Statistics (NCHS) in partnership with the National Association for Public Health Statistics and Information Systems and the

Robert Wood Johnson Foundation. USALEEP was designed to estimate life expectancy at the census-tract level so that researchers can track health disparities between neighborhoods, which county averages conceal. Poverty comes from the Census Bureau's Small Area Income and Poverty Estimates (SAIPE) program, which produces timely, model-based estimates of income and poverty for counties and school districts between decennial censuses, primarily to guide the allocation of federal funds. Educational attainment and the urban–rural code come from the USDA Economic Research Service (ERS), which compiles county-level socioeconomic data to support rural development and agricultural policy research. Finally, the 2010 Decennial Census provides a complete count of population and housing units down to the tract, which I use to weight tracts when aggregating to counties and to identify which tracts are missing from USALEEP.

None of the variables is a direct observation of the quantity the model requires, and each reflects decisions made by its producers. Tract life expectancy is a modeled estimate based on deaths over a multi-year window rather than an observed lifespan. I aggregated it to the county as a population-weighted mean of tract values, using 2010 tract populations as weights. The choice is consequential. In some counties the weighted and simple means differ by as much as 3.08 years. SAIPE poverty is likewise a model-based estimate, not a head count. The variables also describe different moments: education refers to 2008–12, population to 2010 and poverty to 2015, so the analysis assumes that county characteristics are stable enough over that period for the measures to be treated as contemporaneous. Metro status is derived from the ERS urban–rural code and is a 0/1 summary of a continuous gradient. The unit of observation is the county, so every conclusion concerns counties and not the individuals who live in them.

Of the 73,057 tracts in the 2010 population file, 5,909 have no USALEEP estimate, and 526 of these had a recorded population of zero. At the county level, 34 SAIPE counties have no USALEEP tract at all, a pattern consistent with small and sparsely populated places. One SAIPE county has a blank poverty value, and 20 ERS county rows are blank on education and the urban–rural code. NCHS suppresses estimates for tracts with very small populations or too few deaths over the observation window, both to avoid statistically unstable figures and to protect individual privacy, and the ERS and SAIPE gaps arise from boundary discrepancies and from counties too small to model reliably. After the merges, 3,108 counties remain, and none has a missing value on any analysis variable, so no imputation or further deletion was required.

The unmerged tracts are not a random subset. They are smaller than the tracts that appear in USALEEP, which rules out Missing Completely at Random. The principal driver of suppression, population size, is observed for every tract in the 2010 file, and it is not the unobserved life expectancy of the suppressed tracts. Conditional on observed population, missingness is therefore plausibly unrelated to the value that is missing, which is the definition of MAR. However the label also depends on the number of deaths, and death counts are themselves related to mortality. Some Missing Not at Random (MNAR) element cannot be excluded. The main implication is that the results describe the 3,108 counties with usable estimates and underrepresent the smallest counties, so they should not be extrapolated to very small or very sparsely populated places.

Table 1. Summary statistics

Variable (units)	Mean	Median	SD	Min	Max	N
Life expectancy at birth (years, population-weighted)	77.8	77.8	2.6	68.9	89.5	3,108
Poverty rate, all ages (%)	16.3	15.2	6.5	3.4	47.4	3,108
Adults 25+ without a high school diploma (%)	15.9	14.5	7.0	2.5	55.1	3,108
Metro county (0/1)	0.4	0.0	0.5	0	1	3,108
Census tracts in the county (count)	21.6	6.0	68.0	1	2,195	3,108
Population, 2010 (people)	93,423	25,716	294,953	763	9,410,747	3,108

Note: Life expectancy is from USALEEP tract estimates, population-weighted to counties with 2010 Census tract populations. Poverty is Census SAIPE, 2015. Adults without a high school diploma is USDA ERS, 2008–12. Metro status is derived from the USDA ERS urban–rural code; its mean is the share of metro counties (about 40%). Tract counts and population are from the 2010 Census. One row is one county.

2. Descriptive Evidence

Starting with the distributions of the outcome and the main explanatory variable. County life expectancy is close to symmetric. Its skewness is 0.25, and its mean and median coincide at 77.8 years (Figure 1, left). Because the outcome is not skewed, there is no case for a log transformation or a median-regression check, and I model life expectancy in levels so that coefficients retain a direct interpretation in years. Poverty is right-skewed (skewness 1.07), with a mean of 16.3% above its median of 15.2% and a long tail reaching 47.4% (Figure 1, right). Skew in an explanatory variable does not violate the OLS assumptions, but it implies that a small number of high-poverty counties may exert high leverage, which I examine in Section 4.

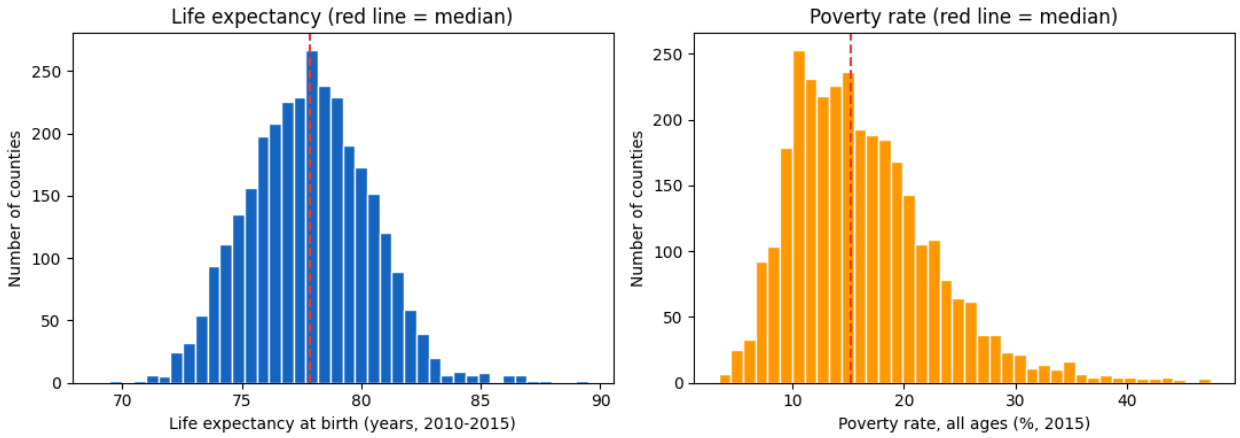


Figure 1. Life expectancy is roughly symmetric around its median of 77.8 years, while poverty piles up at low rates and has a long right tail (median 15.2%, mean 16.3%). Red dashed lines mark the medians. Source: USALEEP and Census SAIPE 2015, 3,108 counties.

The bivariate relationship is strongly negative. The correlation between poverty and life expectancy is −0.58, and a line fitted by OLS through the 3,108 counties has a slope of −0.232 years per percentage

point of poverty (Figure 2). A 10-point difference in poverty is therefore associated with a difference of about 2.3 years of life expectancy, and poverty alone accounts for 34% of the variation in county life expectancy ($R^2 = 0.340$). The scatter around the line is nonetheless wide: counties with similar poverty rates differ in life expectancy by several years, which indicates that factors other than poverty matter and motivates the controls introduced in Section 3.

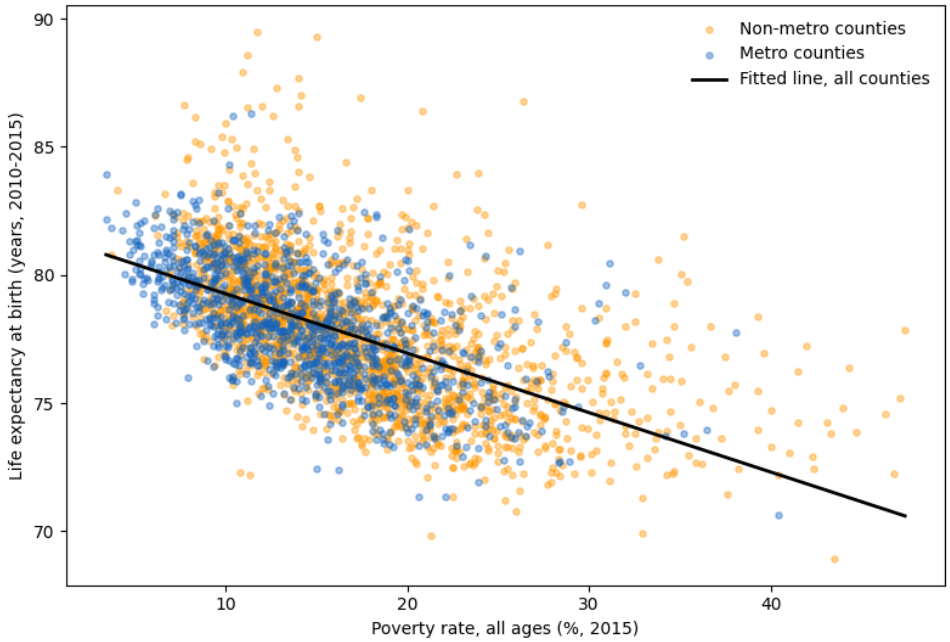


Figure 2. Counties with higher poverty have lower life expectancy: the correlation is −0.58 and the fitted line falls 0.232 years per percentage point. Non-metro counties (orange) are more spread out and include most of the longest-lived counties. Source: USALEEP and Census SAIPE 2015, 3,108 counties.

Non-metro counties are more dispersed, with the highest life expectancies in the sample concentrated among them. They are also poorer on average. Non-metro counties have a mean poverty rate of 17.4% and a mean life expectancy of 77.7 years, whereas metro counties average 14.4% and 78.0 years. The raw gap in life expectancy between the two groups is therefore only 0.3 years, in favor of metro counties, even though the poverty gap is three percentage points. This is the first sign that the urban–rural divide and poverty are entangled, and it will matter when metro status is interpreted as a control.

These descriptive patterns establish that the association is strong, roughly linear and not driven by an obviously skewed outcome. They do not show that poverty has an independent association with longevity, because the raw slope combines any direct association with the effects of everything correlated with poverty, including education and place of residence. Separating those channels is the purpose of the regression in the next section.

3. The Model

The model relates county life expectancy to the poverty rate, the share of adults without a high school diploma and a metro indicator. It is estimated by ordinary least squares across the 3,108 counties:

$$\textit{Life expectancy} = \beta_0 + \beta_1 \cdot \textit{Poverty} + \beta_2 \cdot \textit{No high school diploma} + \beta_3 \cdot \textit{Metro} + \varepsilon,$$

where life expectancy is measured in years and poverty and the no-diploma share in percentage points. The specification is linear in levels, which is justified by the symmetric distribution of the outcome. Standard errors are heteroskedasticity-robust (HC3) because the constant-variance assumption is

rejected in Section 4. The full results are reported in Table A1, and Section 4 examines the model's diagnostics.

The coefficient on poverty is $\beta_1 = -0.169$ years per percentage point (robust standard error 0.010; 95% confidence interval −0.189 to −0.150). Comparing two counties with the same share of adults without a high school diploma and the same metro status, the county with a poverty rate one percentage point higher is predicted to have a life expectancy about 0.17 years shorter. A 10-point difference in poverty, which is well within the observed range of 3.4% to 47.4%, corresponds to roughly 1.7 years (interval 1.5 to 1.9 years). The estimate is large relative to its precision, and the interval excludes zero by a wide margin. Relative to the poverty-only model, the coefficient falls from −0.232 to −0.169, a reduction of about 27%, so roughly a quarter of the raw association overlaps with education and metro status, while the remaining three quarters persist after those controls.

The coefficient on the share of adults without a high school diploma is $\beta_2 = -0.094$ (standard error 0.010; interval −0.113 to −0.074). Holding poverty and metro status fixed, a county in which ten more percentage points of adults lack a diploma has a life expectancy about 0.94 years shorter. Education is therefore associated with longevity separately from poverty, although its coefficient is about half the size of the poverty coefficient per percentage point.

The coefficient on the metro indicator is $\beta_3 = -0.460$ (standard error 0.074; interval −0.605 to −0.315). Holding poverty and education fixed, metro counties have a life expectancy about half a year shorter than non-metro counties. This contrasts with the raw difference of 0.3 years in favor of metro counties documented in Section 2. The sign reverses because metro counties are less poor, and that advantage is already captured by the poverty term. Once poverty is held constant, the remaining difference reflects other features of metro counties. The intercept (82.2 years) is the prediction for a non-metro county with 0% poverty and 0% of adults lacking a diploma, which lies outside the data (the minimum poverty rate is 3.4%), so I do not interpret it. The model explains 37.9% of the variation in life expectancy, 3.9 percentage points more than poverty alone, which indicates that most of the variation in county life expectancy remains unexplained by these three variables.

The share of adults without a diploma is included to block a confounding path. Counties with less schooling tend to have both higher poverty and worse health behaviors and access to care, so omitting education would attribute some of its association with longevity to poverty. It would be a bad control if it were itself partly a consequence of poverty, because conditioning on a mediator removes part of the very association of interest and would understate poverty's total association. Adult education measured in 2008–12 was largely determined before the 2015 poverty rate, which limits that concern without eliminating it. Metro status is included to block the path through urban–rural differences in labor markets, health care supply and population composition, and to compare like with like. It would be a bad control if poverty caused a county's classification, which is unlikely over this horizon, but as a 0/1 variable it leaves most regional and urban–rural variation unaccounted for, so it improves comparability only partially.

4. Diagnostics

The residuals range from −7.0 to 11.6 years, with a standard deviation of 2.02 years, so a typical county's life expectancy is predicted to within about two years. Figure 3 plots them against fitted values and against the poverty rate. They are centered on zero with a long positive tail, meaning that the model under-predicts the life expectancy of some counties by a large margin. The vertical spread appears wider at low poverty than at high poverty, although there are far fewer high-poverty counties, which is consistent with the formal test below. Above roughly 30% poverty, most residuals are positive, which

suggests that the straight line may overstate the life-expectancy shortfall for the poorest counties. This is a mild sign of nonlinearity that the present specification does not model.

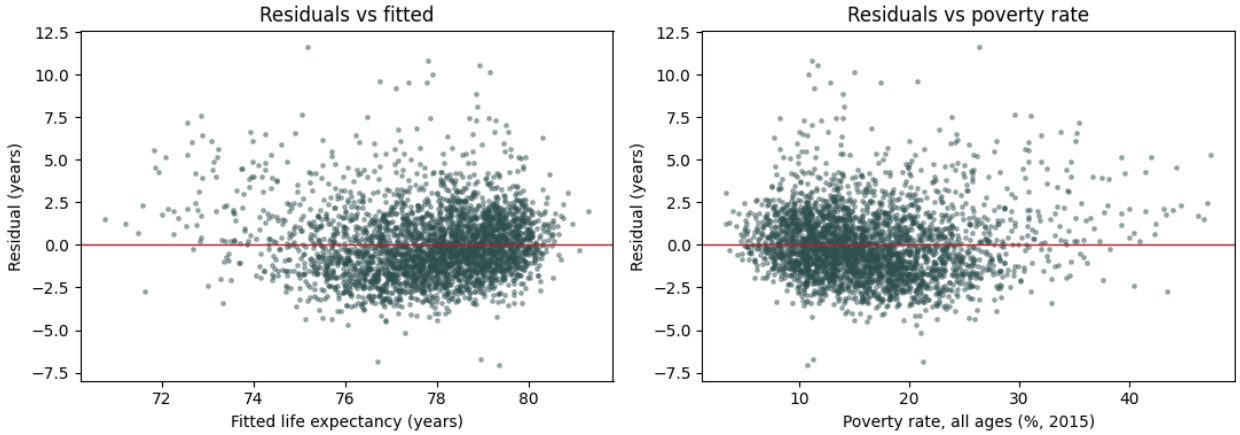


Figure 3. Residuals center on zero with a long positive tail and a spread that varies with poverty and with fitted life expectancy, which is why inference uses robust standard errors (Breusch–Pagan $p < 0.001$). Source: model (2), 3,108 counties.

The Breusch–Pagan test rejects the null of constant variance, so I report HC3 heteroskedasticity-robust standard errors throughout. Relative to the classical standard errors, the robust errors are larger for poverty (0.0100 versus 0.0078, a ratio of 1.29) and for education (0.0099 versus 0.0071, a ratio of 1.39), and about the same for metro status (0.0738 versus 0.0771, a ratio of 0.96). Heteroskedasticity does not bias the coefficients, but it would have made the classical errors for the first two variables too small, and therefore the intervals too narrow. Under either set of errors, the interval for poverty remains far from zero, so the substantive conclusion does not depend on the correction.

The variance inflation factors are 1.93 for poverty, 1.91 for the share without a diploma and 1.06 for metro status. Poverty and education are correlated, as expected, but all three factors are well below the conventional warning level of 5. A VIF of 1.93 means the variance of the poverty coefficient is about 1.9 times what it would be if poverty were uncorrelated with the other regressors, which is modest. Collinearity is therefore not what limits the precision of the estimates.

The largest Cook's distance in the sample is 0.036, far below the conventional threshold of 1. Five counties nonetheless account for 10% of the total Cook's distance. Holmes, Martin, Haskell, Corson and Starr (Table A2). Two of them are extreme on explanatory variables. Corson has the highest poverty rate in the sample (47.4%) and Starr the highest share of adults without a diploma (55.1%). The other three combine unusually long lives (86 to 89 years) with moderate poverty and produce the largest residuals. Such values look more like noisy small area estimates than genuine outliers, although I have not verified this. To assess their effect, I re-estimated the model without all five counties. The poverty coefficient moves from −0.1692 to −0.1645, a change of about 3%, so the main result does not depend on any small group of observations.

5. Threats and Conclusion

For the poverty coefficient to represent a causal effect, counties with different poverty rates would have to be alike in every other determinant of longevity, apart from the education and metro status that are controlled for, and poverty would have to precede and not follow life expectancy. I do not think these conditions hold, for four reasons.

First, the causation plausibly runs in both directions. Illness lowers earnings and raises medical costs, so low life expectancy can push poverty up as well as the reverse, and a cross-section cannot distinguish

the two. Second, important determinants of both variables are omitted. Smoking, obesity, age and racial composition, access to health care and regional factors plausibly raise poverty and shorten life at the same time, which would bias the coefficient away from zero in absolute value. Third, education is an uncertain control, since it may partly be a result of poverty. Controlling for it could understate poverty's total association, so the move from −0.232 to −0.169 shows how much of the raw gap overlaps with education and metro status but says nothing about what would remain after all confounders are removed. Fourth, measurement adds doubt. The SAIPE poverty is a model-based estimate, life expectancy comes from modeled tract estimates averaged to counties, 5,909 tracts and 34 small counties are missing, and county averages say nothing about individuals, so the results are subject to ecological inference. Neighboring counties are also unlikely to be independent, so the standard errors reported here may be too small.

The coefficient could be conditional association. Counties with a poverty rate one percentage point higher, and the same adult education and metro status, have about 0.17 fewer years of life expectancy. It is a descriptive regularity that any causal explanation of county health must be consistent with. It is not an estimate of the gain in life expectancy that would follow from reducing poverty.

The negative association between poverty and life expectancy is large relative to its uncertainty and survives the diagnostics. The variance inflation factors are below 2, the largest Cook's distance is 0.036, and dropping the five most influential counties barely moves the coefficient. Breusch–Pagan rejects constant variance, which is why robust standard errors are reported. So, a reader should place considerable confidence in the sign and rough magnitude of the association but not read it as the effect of reducing poverty. The model explains 38% of the variation in county life expectancy, and the typical residual is about 2 years, so much of the variation remains unexplained.

With another month, I would extend the analysis. I would cluster standard errors by state or add state fixed effects to address dependence between neighboring counties and absorb regional differences. I would add health-behavior and demographic controls to narrow the set of omitted confounders. I would weight counties by pooled population, as USALEEP does, so that estimates for small counties do not carry the same weight as those for large ones. I would investigate the nonlinearity at high poverty, for example with a quadratic term or a spline. Finally, I would use changes in poverty over time instead of a single cross-section, which would speak more directly to the direction of causation.

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Appendix

Table A1. County life expectancy at birth (years) regressed on poverty and controls

	(1) Poverty only	(2) Main, with controls
Poverty rate (%)	−0.232	−0.169
	(0.007)	(0.010)
No high school diploma (%)		−0.094
		(0.010)
Metro county (0/1)		−0.460
		(0.074)
Constant	81.581	82.225
	(0.111)	(0.127)
N	3,108	3,108
R^2	0.340	0.379

Note: Dependent variable: county life expectancy at birth (years). Standard errors in parentheses: HC3 heteroskedasticity-robust. Column 1: poverty only, the raw association. Column 2: adds the share of adults without a high school diploma and a metro indicator. Sources: USALEEP, Census SAIPE 2015, USDA ERS.

Table A2. The five most influential counties (by Cook's distance)

County	Poverty (%)	No HS diploma (%)	Metro	Life exp. (yrs)	Leverage	Std. residual	Cook's D
Holmes County	10.8	44.5	0	82.6	0.014	3.20	0.036
Martin County	11.4	29.3	1	86.3	0.005	4.57	0.025
Haskell County	11.2	27.0	0	88.6	0.003	5.35	0.025
Corson County	47.4	17.6	0	77.9	0.014	2.64	0.024
Starr County	30.9	55.1	0	77.4	0.012	2.76	0.023

Note: Poverty coefficient with all counties: −0.1692; without these five: −0.1645. Source: model (2).

Table A3. Classical and robust standard errors, and variance inflation factors, model (2)

Term	Coefficient	Classical SE	HC3 SE	HC3 / classical	VIF
Constant	82.2254	0.1144	0.1271	1.11	—
Poverty rate (%)	−0.1692	0.0078	0.0100	1.29	1.93
No high school diploma (%)	−0.0937	0.0071	0.0099	1.39	1.91
Metro county (0/1)	−0.4601	0.0771	0.0738	0.96	1.06

Note: Breusch–Pagan p-value 3.58×10^{-22}. Source: model (2), 3,108 counties.